ASTR 1020 Lab

L8: Galaxy Classification



Credit: Ken Crawford/Rancho Del Sol Observatory

Grading

- All labs will be graded out of **10 points***

Points per question	Description
1.0	A correct answer **with** units **and** work shown. Answers that don't require work will be **graded on completion**
0.8	A correct answer **without** units or work shown
0.6	An incorrect answer **with** units **and** work shown
0.4	An incorrect answer **without** units or work shown
0.2	Some work shown **without** an answer
0.0	Not Attempted

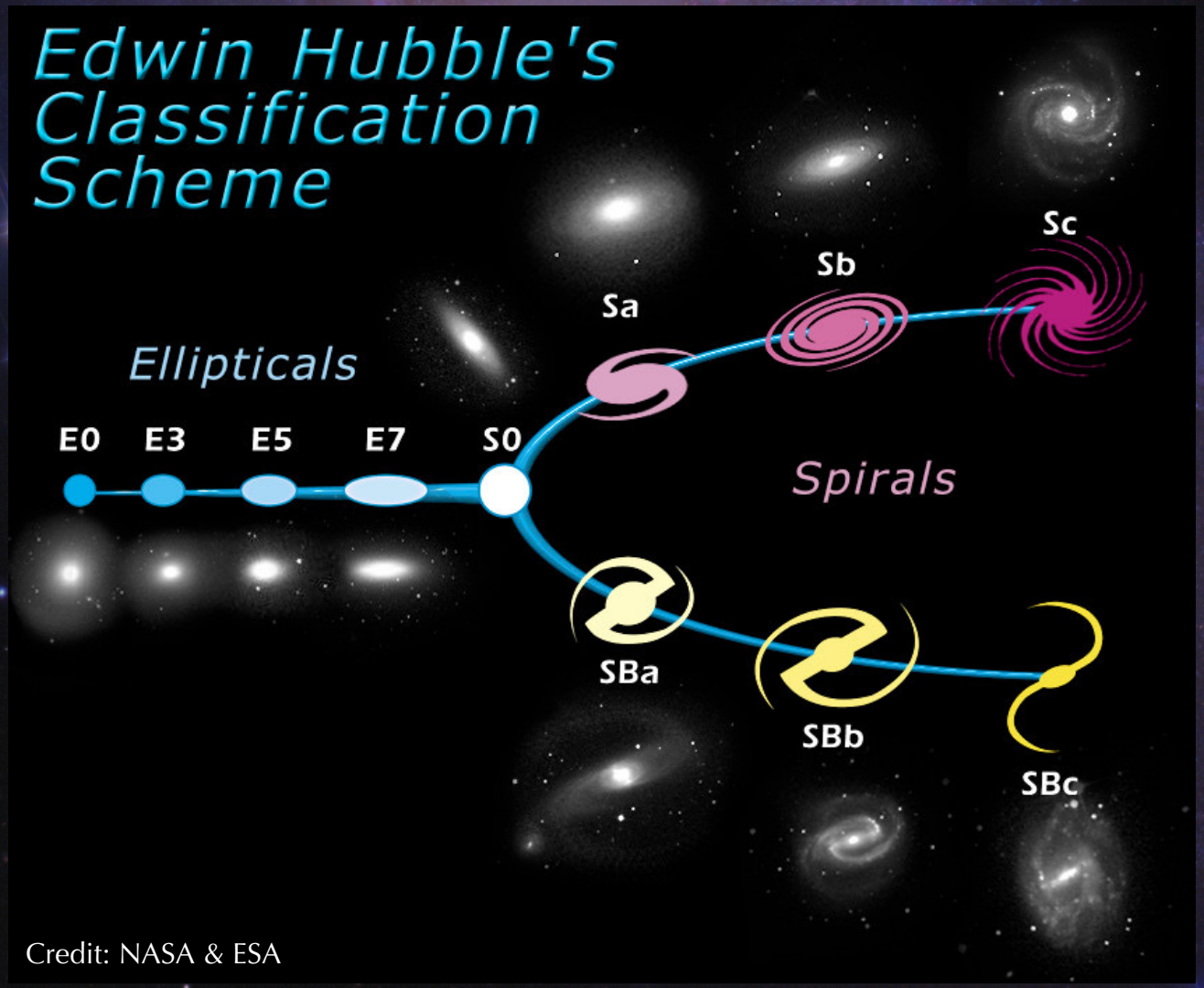
Lab Schedule

Lab	Dates	Topic
~~Lab 1~~	~~September 14~~	~~Excel & Data Analysis~~
~~LAB 2~~	~~September 21~~	~~Sun~~
~~LAB 3~~	~~September 28~~	~~Brightness & Luminosity~~
~~LAB 4~~	~~October 5~~	~~Spectra~~
~~LAB 5~~	~~October 12~~	~~Period-Luminosity~~
~~LAB 6~~	~~October 19~~	~~Hubble~~
~~LAB 7~~	~~October 26~~	~~Stellar Evolution~~
LAB 8	November 2	Galaxies
LAB 9	November 9	Star Clusters & Supernovae
LAB 10	November 16	Black Holes

Things You Need To Know For Lab 8

- The different types of galaxies
- Finding percentages
- The age of the Universe: **13.8 billion years = 1.38×10^{10} years**

Types of Galaxies



Credit: NASA & ESA

Elliptical Galaxies

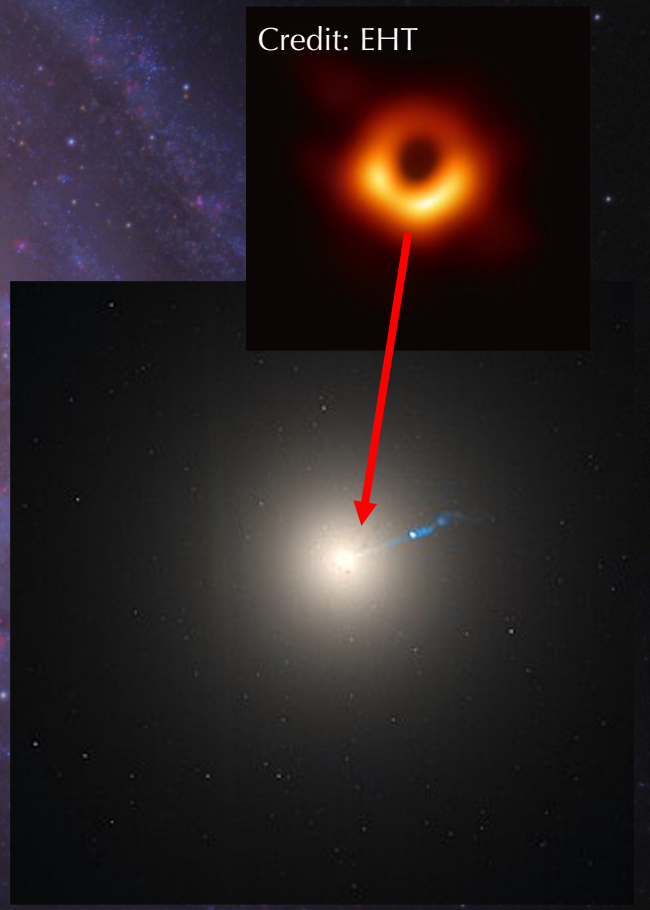
- Old, red stars
- Roughly circular or elongated in shape



Credit: ESA/Hubble



Credit: NASA/ESA/Hubble/J. Blakeslee (WSU)



Credit: NASA/STScI/Hubble

Spiral Galaxies

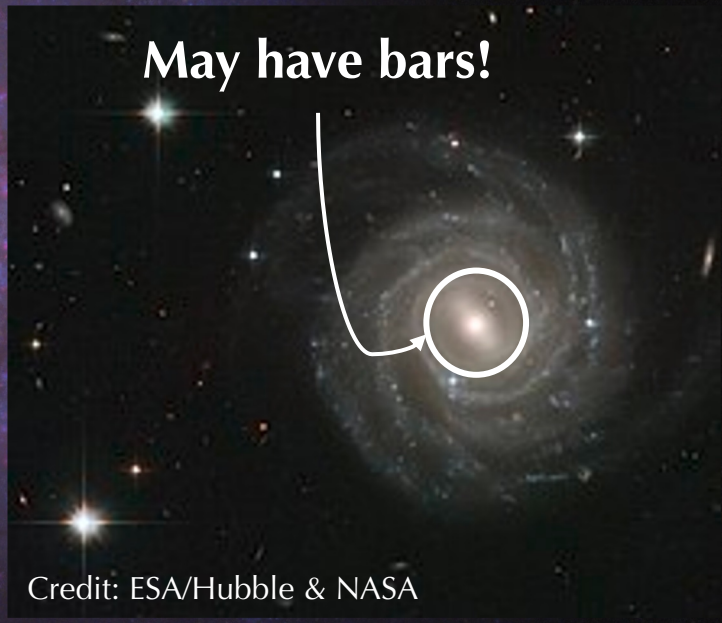
- Younger
- Actively forming stars — look for red or blue fuzzy blobs in arms!



Credit: NASA/ESA/STScI/CFHT/NOAO



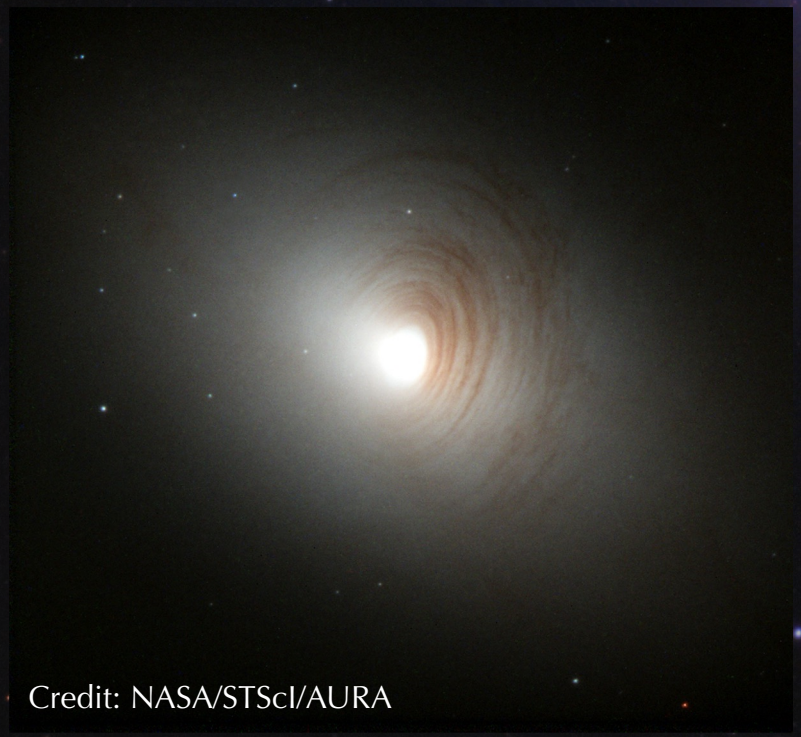
Credit: ESO/VLT



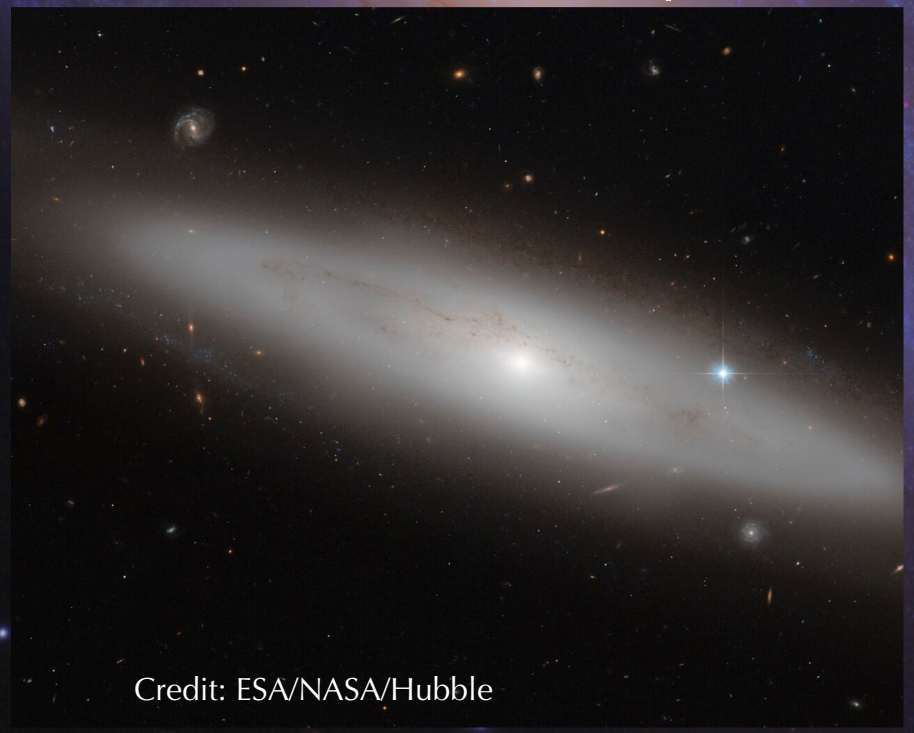
Credit: ESA/Hubble & NASA

Lenticular Galaxies

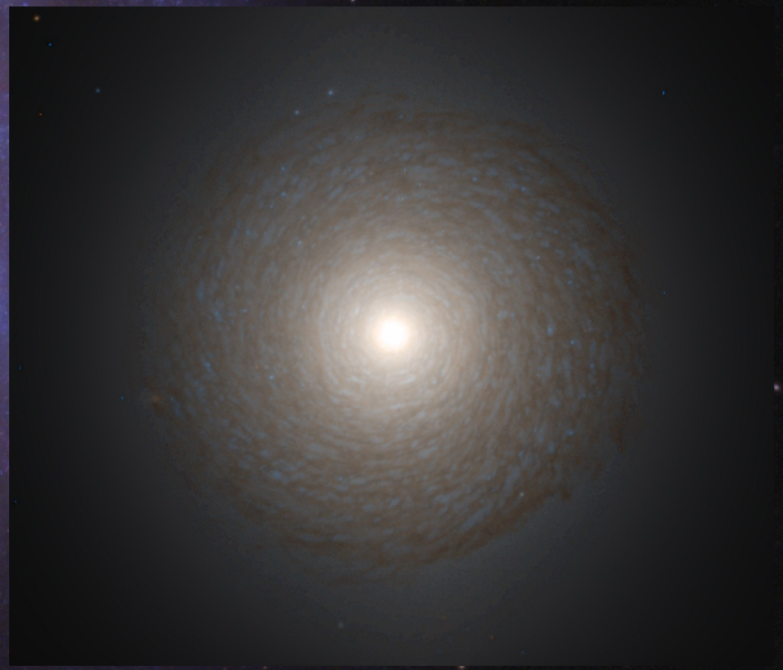
- In-between spirals and ellipticals
- Flat and have a disk like spirals but **lack arms!**
- No active star formation—lack the fuzzy blobs!



Credit: NASA/STScI/AURA



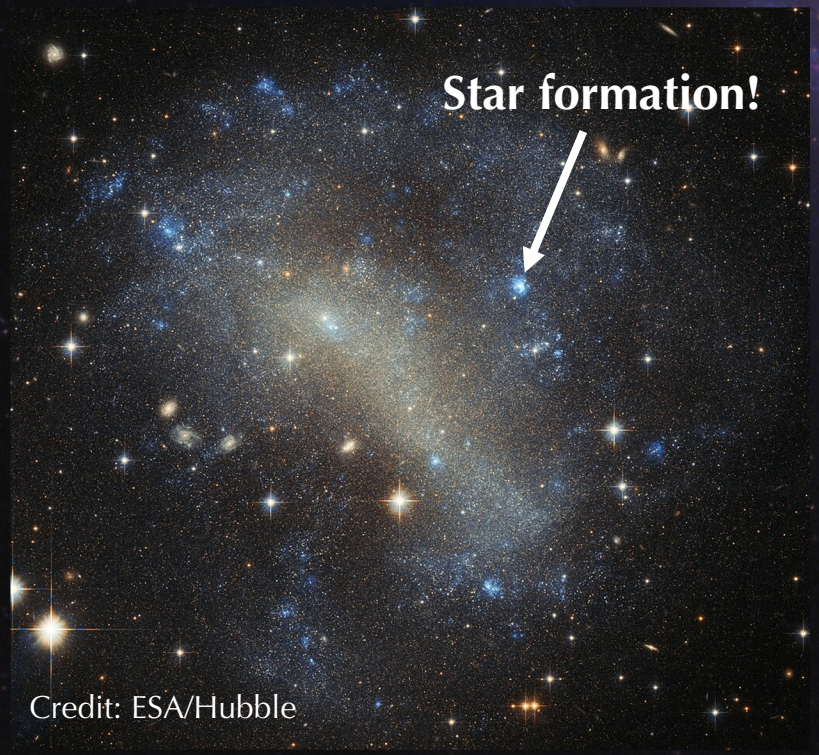
Credit: ESA/NASA/Hubble



Credit: NASA/Hubble

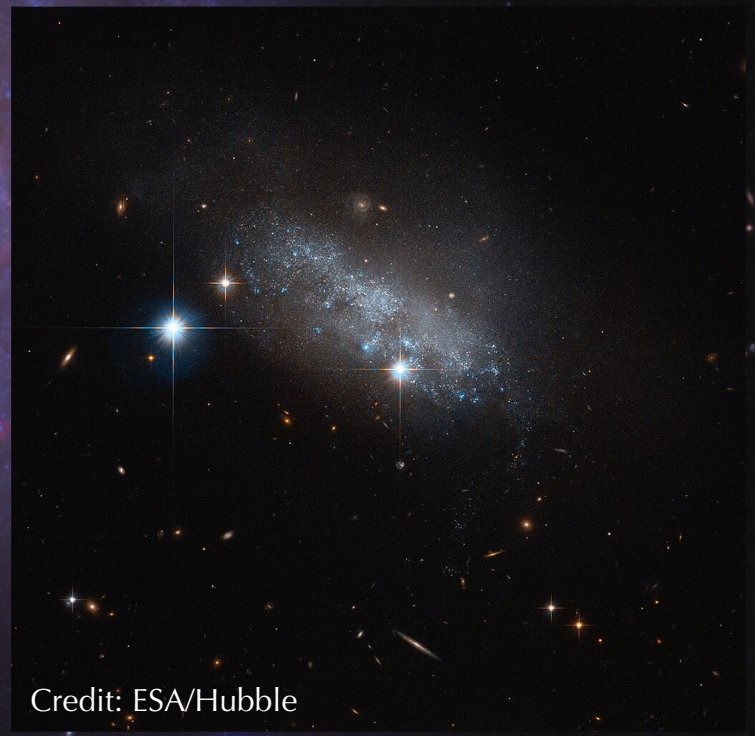
Irregular Galaxies

- Also known as **train-wreck galaxies**
- Thought to form from **mergers and collisions** of other galaxies!



Credit: ESA/Hubble


Credit: NASA/ESA/STScI/AURA


Credit: ESA/Hubble

Finding Percentages: Two Examples

- What is 5% of 30?

$$\frac{5}{100} \times 30 = 1.5$$

- What percentage of 30 is 5?

$$\frac{5}{30} \times 100 = 16.667\%$$

- **Note the difference!**


NGC 4490


UGC 2885


UGC 5189A

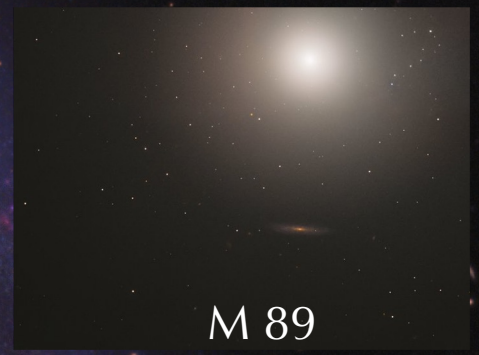

NGC 3059


NGC 1132


NGC 4449


NGC 4621


M 83


M 89


NGC 2985


NGC 6861


NGC 3558 [left]

LEDA 83465 [right]


NGC 6684


NGC 691